From a vague ask to architectural bounds (worked example)

1. What is the true problem?

"our app is slow" → TTFT p99 1.4 s on a 70B model

2. What does success look like?

p95 TTFT < 300 ms, TPOT ~25 ms, 40 rps peak

3. What data exists?

9.2K-token prompts, ~55% share a common prefix

4. What are the constraints?

8×H100 host, ~$14.6K/mo, no SLO headroom

5. What is in scope now?

KV precision FP16→FP8 + prefix cache + routing

Quantified bounds → {TTFT, throughput, cost} budget